

The whitepaper "Agent Quality," authored by Meltem Subasioglu, Turan Bulmus, and Wafae Bakkali, and dated November 2025, presents a framework for evaluating and ensuring the quality of autonomous AI agents in a non-deterministic world.

I. The Paradigm Shift: From Predictable Code to Unpredictable Agents

The **agentic era** represents a profound shift in software engineering, moving from predictable, instruction-based tools to **autonomous, goal-oriented AI agents**. This shift introduces inherent **non-determinism**, which breaks traditional models of quality assurance (QA).

Why Traditional QA Fails:

- **Analogy:** Traditional software is like a predictable delivery truck; an AI agent is a complex, autonomous Formula 1 race car whose success depends on dynamic judgment.
- **Verification vs. Validation:** Traditional software verification asks, "Did we build the product right?" (verifying logic against a fixed spec). Modern AI evaluation is a process of **validation**, asking the more complex question, "Did we build the right product?" (assessing quality in a dynamic world).
- **Agent Failures:** Failures are often subtle degradations of quality rather than system crashes. The output may look plausible but be profoundly wrong and erode trust.
- **Key Failure Modes (Table 1):**
 - **Algorithmic Bias:** Operationalizing and amplifying systemic biases from training data.
 - **Factual Hallucination:** Producing plausible-sounding but factually incorrect or invented information with high confidence.
 - **Performance & Concept Drift:** Agent performance degrades over time as real-world data changes.
 - **Emergent Unintended Behaviors:** Agents develop novel strategies that may be inefficient, unhelpful, or exploitative (e.g., exploiting loopholes or engaging in "proxy wars").

The Architectural Shift to System-Centric AI:

The complexity of evaluation has compounded across stages, moving from Traditional ML to Multi-Agent Systems. Active AI agents break old evaluation models due to three core capabilities:

1. **Planning and Multi-Step Reasoning:** Agents decompose goals, creating a **trajectory** (Thought → Action → Observation → Thought...) where non-determinism compounds at every step.

2. **Tool Use and Function Calling:** Agents interact with the real world via external APIs and tools, introducing dynamic environmental interaction.
3. **Memory:** Agents maintain state (short-term "scratchpad" and long-term memory), meaning behavior evolves, and results can change based on what the agent has "learned".

The ultimate unit of evaluation is no longer the model, but the **entire system trajectory**.

The Four Pillars of Agent Quality (The Goals):

Agent quality must be measured holistically across four interconnected pillars:

1. **Effectiveness (Goal Achievement):** The ultimate "black-box" question: Did the agent accurately achieve the user's intent?. This connects to user-centered metrics and business KPIs.
2. **Efficiency (Operational Cost):** Did the agent solve the problem well? Measured by resources consumed: **total tokens** (cost), **wall-clock time** (latency), and **trajectory complexity** (total steps).
3. **Robustness (Reliability):** How gracefully does the agent handle adversity (e.g., API timeouts, ambiguous prompts)? A robust agent fails gracefully, retries calls, or asks for clarification.
4. **Safety & Alignment (Trustworthiness):** The non-negotiable gate. Ensures the agent operates within defined ethical boundaries, covering Responsible AI metrics for fairness, bias, and security against prompt injection.

II. The Art of Agent Evaluation: Judging the Process

Effective evaluation follows the **"Outside-In" Evaluation Hierarchy**, prioritizing real-world success before examining technical details.

1. The "Outside-In" View (The Black Box)
This is the end-to-end evaluation focusing on the agent's final performance against its defined objective.
 - **Metrics:** Task Success Rate (e.g., successful transaction rate), User Satisfaction (CSAT, thumbs up/down), and Overall Quality (accuracy, completeness).
2. The "Inside-Out" View (The Glass Box)
When the final output is flawed, this view systematically assesses every component of the execution trajectory. Diagnostic power is achieved by moving from "the final answer is wrong" (Black Box) to "the final answer is wrong because..." (Glass Box).

- **LLM Planning (The "Thought"):** Checking for core reasoning failures like hallucinations, context pollution, or repetitive loops.
- **Tool Usage:** Assessing if the agent chose the wrong tool, failed to call a necessary tool, or provided malformed JSON or incorrect parameters.
- **Tool Response Interpretation (The "Observation"):** Determining if the agent misinterpreted numerical data, failed to extract key entities, or critically, failed to recognize and react to an error state (e.g., an API's 404 error).
- **RAG Performance:** Checking for irrelevant document retrieval, outdated information, or the LLM ignoring retrieved context.
- **Trajectory Efficiency and Robustness:** Exposing inefficient resource allocation (e.g., excessive API calls) and unhandled exceptions.
- **Multi-Agent Dynamics:** In advanced systems, checking inter-agent communication logs for misunderstandings or adherence to defined roles.

The Evaluators: A Hybrid Approach:

Judgment requires a sophisticated mix of automated scalability and human nuance.

Evaluator Type	Description	Key Application
Automated Metrics	String-based (ROUGE, BLEU) and embedding-based (BERTScore) similarity scores.	Efficient, low-cost "first filter" for regression testing and detecting obvious failures at scale.
LLM-as-a-Judge	Uses a powerful, state-of-the-art model (like Gemini Advanced) to evaluate the qualitative outputs of another agent based on a detailed rubric.	Provides scalable, fast, nuanced feedback for outputs and intermediate steps. Pairwise comparison is recommended to mitigate biases and reliably signal improvement.
Agent-as-a-Judge	Uses a specialized "Critic Agent" to evaluate the full execution trace of another agent.	Valuable for process evaluation , assessing plan quality, correct tool selection, and context handling, detecting inefficient process failures.
Human-in-the-Loop (HITL)	Indispensable for capturing critical qualitative signals, complex domain knowledge, and subtle nuances (tone, creativity).	Establishes the "gold standard" benchmark and creates the comprehensive "Golden Set" of test cases.

User Feedback & Reviewer UI	Captures real-world signals (e.g., thumbs up/down, PR acceptance rate) and presents them alongside the agent's reasoning trace.	Enables governance dashboards and supports interruption workflows for high-stakes tool calls, requiring human approval before proceeding.
--	---	--

Responsible AI (RAI) and Safety Evaluation:

This is a **mandatory, non-negotiable gate**. Evaluation involves:

- **Systematic Red Teaming:** Actively trying to break the agent using adversarial scenarios (e.g., generating hate speech, prompt injection).
- **Adherence to Guidelines:** Explicitly evaluating outputs against predefined ethical guidelines and principles.
- **Guardrails Architecture:** Guardrails should be implemented as a structured Plugin with internal methods registered to framework callbacks (e.g., `check_input_safety` using `before_model_callback`).

III. Observability: Seeing Inside the Agent's Mind

Observability is the framework that provides the crucial insight needed to trust an autonomous agent. The goal is to move from simple monitoring to true observability—understanding the quality of the agent's cognitive processes ("Is the agent thinking effectively?").

The Three Pillars of Observability

These pillars provide the "eyes and ears" inside the agent's process:

1. Pillar 1: Logging – The Agent's Diary:

- **What it is:** Atomic, timestamped entries representing raw, immutable facts (what happened).
- **Best Practice:** Use structured JSON format to capture rich context: prompt/response pairs, intermediate reasoning steps (Chain of Thought), structured tool calls, and state changes.
- **Tradeoff:** Verbose DEBUG logs are useful for development, but dynamic filtering is needed in production.

2. Pillar 2: Tracing – Following the Agent's Footsteps:

- **What it is:** The narrative thread that connects individual logs (spans) into a coherent, end-to-end view of a single task. Traces reveal the causal chain (the crucial "**why**").

- **Components:** Built on standards like OpenTelemetry. A trace links Spans (individual operations, e.g., llm_call), Attributes (metadata like token_count, latency_ms), and uses Context Propagation (unique trace_id).

3. Pillar 3: Metrics – The Agent's Health Report:

- **What it is:** Quantitative, aggregated scores derived from logs and traces over time (how well it happened, on average?).
- **System Metrics (Vital Signs):** Foundational, quantitative measures of operational health for SREs/DevOps.
 - *Examples:* P50/P99 Latency, Error Rate, Cost (Tokens per Task, API Cost per Run), and Tool Usage Frequency.
- **Quality Metrics (Judging Decision-Making):** Second-order metrics derived by applying judgment frameworks on top of raw data, assessing the agent's reasoning and final output.
 - *Examples:* Correctness & Accuracy, Trajectory Adherence (following the ideal path), Safety & Responsibility, and Helpfulness & Relevance.

Operational Insights:

- **Dashboards:** Requires distinct **Operational Dashboards** (tracking latency, error rate) and **Quality Dashboards** (tracking factual correctness score, helpfulness) to serve different teams (SREs vs. Data Scientists).
- **Granularity vs. Overhead:** Managed using **Dynamic Sampling**—logging detailed information (DEBUG level) in development, but setting a lower default in production (INFO), while tracing 100% of errors but only a sample (e.g., 10%) of successful requests.
- **Security:** PII scrubbing mechanisms must be integrated into the logging pipeline before data is stored.

IV. The Agent Quality Flywheel (Conclusion)

The Agent Quality Flywheel is the operational playbook for continuous evaluation and improvement, turning principles into a self-reinforcing system.

Steps of the Flywheel:

1. **Step 1: Define Quality (The Target):** Set concrete targets using the **Four Pillars** (Effectiveness, Efficiency, Safety, Trust).
2. **Step 2: Instrument for Visibility (The Foundation):** Generate the evidence (Logs and Traces) by adhering to the Observability principles.

3. **Step 3: Evaluate the Process (The Engine):** Judge performance using the "Outside-In" assessment, utilizing scalable LLM-as-a-Judge systems and HITL judgment.
4. **Step 4: Architect the Feedback Loop (The Momentum):** Programmatically convert every production failure, captured and annotated, into a permanent regression test in the "Golden" Evaluation Set, driving continuous improvement.

Three Core Principles for Building Trustworthy Agents:

1. **Treat Evaluation as an Architectural Pillar, Not a Final Step:** Agents must be "evaluatable-by-design," instrumented from the start to emit the necessary logs and traces.
2. **The Trajectory is the Truth:** The true measure of quality lies in the agent's end-to-end "thought process" (Process Evaluation), which requires deep Observability.
3. **The Human is the Arbiter:** While automation is crucial for scale, the final definition of "good," along with judgments on safety and fairness, must be anchored to human values through **HITL evaluation**.

Mastering this discipline of "**Evaluation Engineering**" is necessary to move beyond capable agents to truly reliable and trustworthy enterprise-grade AI systems.

Analogy: If evaluating agent quality were like cooking a meal, **Observability** provides the infrastructure—the camera angles, temperature gauges, and recipe book—to see *how* the gourmet chef made the dish. **Evaluation** is the food critic's job: tasting the final dish (Black Box) but also reviewing the prep notes and techniques (Glass Box) to understand *why* the dish succeeded or failed, judging it not just by taste (Effectiveness) but also by efficiency and safety. The **Agent Quality Flywheel** is the business process that ensures every time the chef fails, the critic's notes are incorporated to make the next dish better.